



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN GEOGRAPHY AND
NATURAL RESOURCES MANAGEMENT WITH INFORMATION
TECHNOLOGY**

MAIN CAMPUS

NGE 104: FUNDAMENTALS OF PHYSICAL GEOGRAPHY

Date: 2nd July, 2018

Time: 8.30- 11.30 am

INSTRUCTIONS

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate.



NGE 104 FUNDAMENTALS OF PHYSICAL GEOGRAPHY

- 1 a) Explain the vertical layering of the atmosphere. (10 marks)
b) Write explanatory notes on the following.
 - i. Isostasy. (5 marks)
 - ii. Carbon Assimilation (5 marks)
 - iii. Intertropical Convergence Zone (5 marks)
 - iv. Volcanism (5 marks)
2. Discuss the factors that influence regional climate. (20 marks)
3. Describe the formation of the Rift Valley. (20 marks)
4. Account for the distribution of species in the biosphere. (20 marks)
5. "The present world continents were once one land mass". Discuss. (20 marks)
6. Discuss the role of surface water in socio-economic development. (20 marks)



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OF EDUCATION WITH INFORMATION TECHNOLOGY**

CITY CAMPUS

NGE 104: FUNDAMENTALS OF PHYSICAL GEOGRAPHY

Date: 13th July , 2018

Time: 9.00 - 12.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.
- Sketch maps and diagrams should be used whenever appropriate



NGE 104 FUNDAMENTALS OF PHYSICAL GEOGRAPHY

- 1 a) Explain the evidence of the continental drift theory. (10 marks)
- b) Examine the constructive impact of vulcanicity. (10 marks)
- c) Explain the ecological processes which regulate and sustain ecosystem functions. (10 marks)
2. Discuss the significance of erosion as a denudation process. (20 marks)
3. Account for the variation of temperature in the atmosphere. (20 marks)
4. Discuss the factors which influence the pattern of regional climates. (20 marks)
5. "Geophysical factors are critical in explaining the temporal and spatial distribution of biota in the ecosphere". Discuss. (20 marks)
6. Discuss the concept of hydrological cycle and explain its role in water resource pollution. (20 marks)



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TECHNOLOGY**

CITY CAMPUS

NGE 105: SPATIAL ORGANIZATION

Date: 11th July, 2018

Time: 2.00 - 5.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used whenever appropriate



NGE 105 SPATIAL ORGANIZATION

- 1 a) Describe the various phases in the development of geography as a science. (10 marks)

b) Distinguish between Inductive and Deductive verification in geography. (10 marks)

c) Explain the barriers to spatial diffusion. (10 marks)
2. a) Distinguish between positivism and critical theory. (5 marks)
b) Differentiate between humanistic and structuralism approaches to geographic research. (15 marks)
3. Examine the economic argument for and against Von Thunen's model in explaining agricultural land use in Kenya. (20 marks)
4. Discuss the relevance of Weber's Industrial Location Theory to Kenya. (20 marks)
5. Describe the Central Place Theory. (20 marks)
6. Assess the extent to which the multiple nuclei model explains the lay out of Nairobi City. (20 marks)



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MAIN CAMPUS

NGE 105: SPATIAL ORGANIZATION

Date: 7th July, 2018

Time: 8.30 – 11.30 am

INSTRUCTIONS

- Answer question ONE and any other TWO questions
- Sketch maps and diagrams should be used whenever appropriate



NGE 105 SPATIAL ORGANIZATION

- 1 a) Describe the sub-divisions of human geography. (8 marks)
- b) Explain the evolution of human geography. (12 marks)
- c) Explain any five (5) characteristics of spatial organization. (10 marks)
2. Explain Heinrich Von Thunnen's Agricultural Location model. (20 marks)
3. Discuss the relevance of Alfred Weber's Least Cost model to the location of industries in Kenya. (20 marks)
4. Examine the relevance of William Alonso's Urban Land Use Theory to urban land uses in Kenya. (20 marks)
5. Explain the types of spatial diffusion and discuss their barriers. (20 marks)
6. "Complementarity is the foundation of spatial interaction". Discuss. (20 marks)



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MAIN CAMPUS

NGE 106: SURVEYING

Date: 4th July, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used whenever appropriate



NGE 106 SURVEYING

- 1.a) Explain the surveying process. (6 marks)
- b) Discuss the importance of aerial photographs and topographic maps to a surveyor. (8 marks)
- c) Distinguish between the following concepts as used in surveying:
- (i) 100metres square and 100square metres (4 marks)
 - (ii) Polar and Rectangular corodinales (4 marks)
 - (iii) Foward and Back bearing (4 marks)
 - (iv) True North and Magnetic North (4 marks)
2. Examine the role of surveying in the development of devolved governance. (20 marks)
3. a) Write explanatory notes on the following:
- i. Local attraction (7 marks)
 - ii. Magnetic declination (7 marks)
- b) The magnetic bearing of a line was found to be N60030'W in 1990, when the declination was 5010'E. Find its present magnetic bearing if declination is 3000'W. (6 marks)
4. a) Two points are visible on topographic map and an aerialphotograph of the same area. The distance between them on the map is 10cm while on the photograp it is 6cm. Computethe scale of the aerial photograph if the scale of the map is 1:20,000. (10 marks)
- b) Identify large scale and small scale in (a) above. Explain youe answer. (4 marks)
- c) Explain the types of scales used in surveying. (6 marks)
5. The following staff readings were taken during a leveling exercise.
- 2.530, 1.347, 0.823, -2.920, 0.975, -3.041, 1.270, -2.676, 1.421, -2.281, 2.513, 1.559, 0.291 and 1.241.
- The 1st reading was taken with staff held on a temporary Bench Mark (BM). The instrument was shifted after the 5th, 9th and 12th readings. The reading was taken when the staff was held on a BM or RL=126.054M.
- a) Rule out a level bookpage and reduce the levels by the rise and fall method applying the necessary mathematical checks. (16 marks)

- b) Citing examples from the recorded data, distinguish between foresight and intermediate sight. (4 marks)

6. The area is required of a strip of land between a straight fence 80m long, and a stream. Given the following offset measurements from the fence to the stream in metres,

Distance (m)	0	10	20	30	40	50	60	70	80
Offset (m)	17.4	28.1	33.8	37.3	38.0	37.3	33.3	29.6	22.0

- a) Compute the area of this strip of land in hectares using a trapezoidal rule? (10 marks)
- b) Explain any four (4) methods of computing area in surveying. (8 marks)
- c) Differentiate between an open and closed traverse. (2 marks)



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MAIN CAMPUS

NGE 107: REMOTE SENSING

Date: 10th July, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used where appropriate.



NGE 107 REMOTE SENSING

1. a) Outline the major milestones in the development of remote sensing during the World War II (1939-1945). (5 marks)
- b) (i) Describe the most favourable atmospheric conditions for remote sensing in the visible portion of the spectrum. (4 marks)
- (ii) Explain why most remote sensing systems avoid detecting and recording wavelengths in the ultra-violet and blue portions of the spectrum. (4 marks)
- c) Define orthophotos and explain their significance. (6 marks)
- d) Explain three (3) categories of image enhancement methods. (6 marks)
- e) Explain the merits and demerits of unsupervised image classification. (5 marks)
2. Discuss the image resolution types in remote sensing. (20 marks)
3. Examine the role of remote sensing and Geographic Information Systems (GIS) in the conservation and management of natural resources. (20 marks)
4. Discuss five (5) elements of image interpretation on aerial photographs. (20 marks)
5. a) Describe the modes of scanning used to acquire multispectral image data. (10 marks)
- b) Explain five (5) advantages of multispectral scanning over photographic systems in image acquisition. (10 marks)
6. Write explanatory notes on the following:
 - a) Rayleigh scattering (4 marks)
 - b) Collateral information (4 marks)
 - c) Active and passive sensors. (4 marks)
 - d) Radiometric and geometric image distortions (4 marks)



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MAIN CAMPUS

NGE 108: CARTOGRAPHY

Date: 5th July, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used whenever appropriate



NGE 108 CARTOGRAPHY

1. a) Describe the tools used in cartography. (10 marks)
b) Explain the sources of data for map preparation. (10 marks)
c) Compare thematic and topographic maps. (10 marks)
2. Discuss the contributions of the Greek philosophers to the development of cartography. (20 marks)
3. a) Explain the processes involved in map making. (10 marks)
b) Write explanatory notes on the following:
(i) Conical projection (5 marks)
(ii) Azimuthal projection (5 marks)
4. Explain the process of symbol design in map preparation. (20 marks)
5. Discuss the processes of analogue and digital map preparation. (20 marks)
6. Examine the methods of relief representation in topographical maps. (20 marks)



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MAIN CAMPUS

NGE 206: AGRICULTURAL GEOGRAPHY

Date: 2nd July, 2018

Time: 12.00- 3.00 pm

INSTRUCTIONS

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 206 AGRICULTURAL GEOGRAPHY

1. a) Assess the relevance of Agricultural Geography to national development. (10 marks)
b) Examine the effect of irrigation on ecology. (10 marks)
c) Explain the ecological approach to agricultural systems. (10 marks)
2. "Boserup's theory defines agricultural activity in the developing countries". Discuss. (20 marks)
3. Discuss the phases of agricultural evolution and development. (20 marks)
4. "Agricultural systems are man-made ecosystems". Discuss. (20 marks)
5. Discuss four (4) factors that hinder agricultural development in Kenya and suggest possible solutions. (20 marks)
6. Examine the effects of agrochemicals and mechanization on ecology. (20 marks)



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CITY CAMPUS – SCHOOL BASED

NGE 206: AGRICULTURAL GEOGRAPHY

Date: 26th September, 2018

Time: 11.00 - 2.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 206 AGRICULTURAL GEOGRAPHY

1. a) Examine the importance of agricultural geography. (10 marks)
b) Explain any four (4) economic factors that influence global agriculture. (8 marks)

c) Account for the emergence of hybridization. (7 marks)
d) Explain two(2) problems facing agricultural development in Kenya. (5 marks)
2. "Agriculture has evolved over time". Discuss. (20 marks)
3. Discuss the relationship between geography and agricultural problems. (20 marks)
4. Explain the diffusion theory of agricultural activity. (20 marks)
5. "Agricultural system is an artificial ecosystem". Discuss. (20 marks)
6. "Simplification and Irrigation are modern agricultural practices that are necessary evils". Discuss. (20 marks)



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INFORMATION TECHNOLOGY**

MAIN CAMPUS

NGE 207: ENVIRONMENT AND DEVELOPMENT IN AFRICA

Date: 5th July, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used whenever appropriate



NGE 207 ENVIRONMENT AND DEVELOPMENT IN AFRICA

1. a) Describe the physiography of Africa. (10 marks)
b) Discuss the environmental factors which constrain the exploitation of marine resources along the coasts of Eastern Africa. (15 marks)
c) Explain any five (5) determinants of economic growth in Africa. (5 marks)
2. Examine the effects of minerals on political and economic development of African countries. (20 marks)
3. "Agricultural development across Sub-Saharan Africa should focus on improved technology". Discuss. (20 marks)
4. "Climate constraints development across Africa". Discuss. (20 marks)
5. Explain the strategies used by African countries to enhance sustainable utilization of water resources. (20 marks)
6. "High population growth rate promotes development in Africa". Discuss. (20 marks)



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TECHNOLOGY**

MAIN CAMPUS

NGE 208: GEOMORPHOLOGY

Date: 9th July, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used where appropriate.



NGE 208 GEOMORPHOLOGY

1. a) Explain the evolutionary approaches in landform genesis. (10 marks)
b) Discuss significance of quantitative analysis in the study of drainage basins. (20 marks)
2. "Some rocks are more resistant to weathering than others". Discuss. (20 marks)
3. Describe the erosional processes and landforms associated with running water. (20 marks)
4. Describe the rapid flowage processes. (20 marks)
5. Describe the aeolian processes and their resultant features. (20 marks)
6. Discuss the evolution of coastal landforms. (20 marks)



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CITY CAMPUS – SCHOOL BASED

NGE 208: GEOMORPHOLOGY

Date: 26th September, 2018

Time: 3.00 - 6.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Sketch maps and diagrams should be used whenever appropriate



NGE 208 GEOMORPHOLOGY

1. a) Distinguish between Catastrophism and Uniformitarianism. (10 marks)
b) "Physiographic processes are responsible for sculpturing the world landscape". Discuss. (20 marks)
2. "Anthropogenic activities are to blame for biological weathering". Discuss. (20 marks)
3. Discuss the techniques used in geomorphological studies. (20 marks)
4. Define mass wasting and examine the slow flowage processes. (20 marks)
5. a) Account for the formation of karst landscape. (10 marks)
b) Examine the significance of karstification as a geomorphic process. (10 marks)
6. Write explanatory notes on the following:
 - a) Graben (5 marks)
 - b) Landslide (5 marks)
 - c) Sand dune (5 marks)
 - d) Exfoliation (5 marks)



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MAIN CAMPUS

NGE 209: AGROFORESTRY SYSTEMS

Date: 11th July, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used where appropriate.



NGE 209 AGROFORESTRY SYSTEMS

1. a) Define Agroforestry and explain its characteristics. (10 marks)

b) Write explanatory notes on the following:
 - (i) Alley cropping (5 marks)
 - (ii) Sylvopasture (5 marks)
 - (iii) Home gardens (5 marks)
 - (iv) Taungya (5 marks)
2. Examine the criteria for classifying agroforestry systems. (20 marks)
3. Explain the viability of improved fallows and shifting cultivation in sub Saharan Africa. (20 marks)
4. Assess the relevance of agroforestry as a land management technique in Kenya. (20 marks)
5. Discuss the constraints in the adoption of agroforestry systems in Africa. (20 marks)
6. "Lack of political good will is an impediment to the global development of agroforestry research". Discuss. (20 marks)



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MAIN CAMPUS

NGE 307: RESEARCH METHODOLOGY

Date: 10th July, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used where appropriate.



NGE 307 RESEARCH METHODOLOGY

1. a) Distinguish between research and research design. (6 marks)
b) Explain any three (3) tools of research. (9 marks)
c) Explain the components of an academic research report. (15 marks)
2. Define research and discuss its importance. (20 marks)
3. "The research process is cyclic". Discuss. (20 marks)
4. Compare survey and experimental research. (20 marks)
5. Explain any four (4) statistical procedures used in inferential statistics. (20 marks)
6. Write explanatory notes on the following:
 - a) Types of data (4 marks)
 - b) Methods of data collection (8 marks)
 - c) Data analysis (8 marks)



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MAIN CAMPUS

NGE 309: BIOGEOGRAPHY

Date: 3rd July, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used whenever appropriate



NGE 309 BIOGEOGRAPHY

1. a) Explain the rationale for studying biogeography. (10 marks)
b) Distinguish between competitive displacement and competitive exclusion. (6 marks)
c) Explain the determinants of diversity index in isolated ecosystems. (14 marks)
2. Discuss the impact of anthropogenic activities on the natural ecosystems. (20 marks)
3. "The existence of policy, institutional and legal frameworks is not sufficient in the conservation and management of biodiversity in the developing countries". Discuss. (20 marks)
4. Discuss the seres that occur between a mined surface and climax community. (20 marks)
5. Examine the adaptive traits of ASAL biota. (20 marks)
6. "Biological factors are critical in explaining the temporal and spatial distribution of species in the ecosphere". Discuss. (20 marks)



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MAIN CAMPUS

NGE 310: RURAL SETTLEMENTS AND LAND USE

Date: 2nd July, 2018

Time: 12.00- 3.00 pm

INSTRUCTIONS

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 310 RURAL SETTLEMENTS AND LAND USE

1. a) Explain the characteristics and functions of rural settlements in Africa.
(12 marks)
- b) Explain the following concepts:
 - (ii) Growth centres (5 marks)
 - (iii) Service centres (5 marks)
- c) Evaluate the characteristics of rural markets in Africa. (8 marks)
2. "Infrastructure constraints rural development". Discuss. (20 marks)
3. Examine the problems facing Africa's agricultural production systems.
(20 marks)
4. Assess the relevance of non-farm activities to poverty alleviation in Kenya.
(20 marks)
5. Explain the importance of land tenure when designing development projects
in rural communities. (20 marks)
6. Discuss the effectiveness of devolved governments in enhancing rural
development in Kenya. (20 marks)



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MAIN CAMPUS

NGE 311: NATURAL HAZARDS

Date: 7th July, 2018

Time: 12.00 – 3.00 pm

INSTRUCTIONS

- Answer question ONE and any other TWO questions
- Sketch maps and diagrams should be used whenever appropriate



NGE 311 NATURAL HAZARDS

- 1 a) Distinguish between hazard and disaster. (5 marks)
b) Explain any five (5) categories of natural hazards. (10 marks)
c) Examine the phases in controlling and managing natural disaster. (15 marks)
2. Discuss the causes and impact of the recent landslides and mudflows in East Africa. (20 marks)
3. "Climate change is the major cause of the current floods in Kenya". Discuss. (20 marks)
4. Evaluate the impact and response to the current geophysical hazards in Africa. (20 marks)
5. Examine the control and management of the current Ebola Virus in Eastern Africa. (20 marks)
6. Assess the role of Red Cross in disaster management. (20 marks)



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MAIN CAMPUS

NGE 312: GEOGRAPHY OF TOURISM

Date: 5th July, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS

- Answer question ONE and any other TWO questions
- Sketch maps and diagrams should be used whenever appropriate



NGE 312 GEOGRAPHY OF TOURISM

1. a) Explain the development of geography of tourism. (10 marks)
b) "Environmental factors are responsible for spatial movement of tourists".
Discuss. (20 marks)
2. Discuss the impact of tourism on the physical environment. (20 marks)
3. Examine different forms of tourism in East Africa. (20 marks)
4. Discuss conservation measures that could be undertaken to protect tourist sites in the developing countries. (20 marks)
5. "Political instability is a major hindrance to tourism development in Africa".
Discuss. (20 marks)
6. Discuss the impact of tourism on socio-economic development in Kenya. (20 marks)



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MAIN CAMPUS

NGE 313: SOIL AND WATER MANAGEMENT

Date: 2nd July, 2018

Time: 3.30- 6.30 pm

INSTRUCTIONS

- Answer question ONE and any other TWO questions
- Sketch maps and diagrams should be used whenever appropriate



NGE 313 SOIL AND WATER MANAGEMENT

1. a) Write explanatory notes on the following:
 - (i) Pedology and Edaphology (4 marks)
 - (ii) Primary and Secondary minerals (4 marks)
 - (iii) Soil texture (4 marks)
 - (iv) Controls of precipitation interception (4 marks)b) Explain five (5) soil conservation measures used in Kenya. (14 marks)

2. Discuss the characteristics and distribution of any two (2) of the following soil orders. (20 marks)
 - a) Oxisols
 - b) Histosols
 - c) Mollisols

3. Discuss the classes of the soil forming processes. (20 marks)

4. Discuss the key dimensions in the development and management of water resources. (20 marks)

5. a) Examine the forces responsible for water retention in the soil. (6 marks)
b) Describe the soil horizons of moist forest climates. (14 marks)

6. a) Distinguish between point and non-point sources of water pollution. (5 marks)
b) Explain any five (5) effects of water pollution on socio-economic development. (5 marks)
c) Examine the problems of groundwater management. (10 marks)



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MAIN CAMPUS

NGE 315: SPATIAL ANALYSIS USING GIS

Date: 11th July, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Sketch maps and diagrams should be used where appropriate.



NGE 315 SPATIAL ANALYSIS USING GIS

- 1 a) Explain critical aspects of geographic data that differentiates them from other types of data. (6 marks)
b) Define spatial autocorrelation and explain its importance in spatial analysis using GIS. (4 marks)
c) Describe the procedure you would follow in georeferencing a raster image for your spatial analysis project. (20 marks)
2. Examine any five (5) key functions of GIS that are essential when using GIS for spatial analysis. (20 marks)
3. As the managing director of your local electric company, describe how you would apply geometric networks in the management of your electricity network. (20 marks)
4. a) Explain the operations involved in spatial analysis using GIS. (9 marks)
b) Examine the use of spatial statistics in identifying patterns and relationships in spatial analysis using GIS. (11 marks)
5. a) Examine the advantages and disadvantages of using vector data representation in spatial analysis using GIS. (10 marks)
b) Describe how the 'Project' and 'Clip' tools found in Arc GIS 10.4 can be used in geoprocessing for spatial analysis using GIS. (10 marks)
6. a) Discuss neighbourhood operations in spatial analysis using GIS. (12 marks)
b) Examine the data measurement scale used for grouping data for use in spatial analysis using GIS. (6 marks)



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INFORMATION TECHNOLOGY**

MAIN CAMPUS

NGE 409: URBAN GEOGRAPHY

Date: 26th September, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 409 URBAN GEOGRAPHY

1. a) Explain the significance of urbanization. (8 marks)
b) Explain the Positivism and Neoclassical approaches to urban geography. (10 marks)
c) Discuss any three (3) problems associated with urbanization. (12 marks)
2. Discuss the structural analysis approach to urban geography. (20 marks)
3. Explain any five (5) social dimensions of modern urbanization. (20 marks)
4. In reference to any city in Kenya, discuss the concentric ring model of urban growth. (20 marks)
5. Define urban planning and discuss the blue print and participatory theories of urban planning. (20 marks)
6. Write explanatory notes on the following:
 - a) Cities of the industrial era (5 marks)
 - b) History and development of urban geography (5 marks)
 - c) Synoptic theory of urban planning (6 marks)
 - d) Urban politics (4 marks)



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

**FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF SCIENCE IN GEOGRAPHY
AND NATURAL RESOURCES MANAGEMENT AND BACHELOR
OF EDUCATION WITH INFORMATION TECHNOLOGY**

CITY CAMPUS – SCHOOL BASED

NGE 409: URBAN GEOGRAPHY

Date: 26th September, 2018

Time: 3.00 - 6.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Sketch maps and diagrams should be used whenever appropriate



NGE 409 URBAN GEOGRAPHY

1. a) Justify the study of urban geography. (10 marks)
b) Citing examples, account for the emergence of urban centres. (20 marks)
2. Discuss the relevance of Multiple Nuclei model to the growth of Kisumu City. (20 marks)
3. Discuss the role of town/city in enhancing national development. (20 marks)
4. Examine the urban locational conflicts and account for their occurrence. (20 marks)
5. Discuss the relevance of urban planning in the developing countries. (20 marks)
6. Write explanatory notes on the following:
 - a) Urban Primacy (5 marks)
 - b) Zoning (5 marks)
 - c) Urban Sprawl (5 marks)
 - d) Centrality (5 marks)



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UNIVERSITY EXAMINATIONS 2018/2019

**FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF EDUCATION AND
BACHELOR OF SCIENCE IN GEOGRAPHY AND NATURAL
RESOURCES MANAGEMENT WITH INFORMATION
TECHNOLOGY**

CITY CAMPUS – SCHOOL BASED

NGE 410: GEOGRAPHY OF AFRICA

Date: 26th September, 2018

Time: 11.00 - 2.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate.



NGE 410 GEOGRAPHY OF AFRICA

1. a) Account for the location of the continent of Africa. (10 marks)

 b) Examine the spatial characteristics of African states. (20 marks)
2. “African continent has diverse physical landscapes”. Discuss. (20 marks)
3. Examine the socio-economic and political impact of transboundary
 resources in Africa. (20 marks)
4. Discuss the effects of African unity on trade and suggest possible
 solutions. (20 marks)
5. Discuss the dimensions of Africa’s agricultural crisis. (20 marks)
6. Discuss the potential of industrial growth in Africa. (20 marks)



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UNIVERSITY EXAMINATIONS 2018/2019

**FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF SCIENCE IN GEOGRAPHY
AND NATURAL RESOURCES MANAGEMENT AND BACHELOR
OF EDUCATION WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

NGE 410: GEOGRAPHY OF AFRICA

Date: 28th September, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 410 GEOGRAPHY OF AFRICA

1. a) Define environmental hazards and explain their effect on development in Africa. (10 marks)

 b) Discuss the geographical basis of African unity. (20 marks)
2. Discuss the impact of rapid population growth on Africa's development. (20 marks)
3. Examine the spatial characteristics of African states. (20 marks)
4. Describe the modified climate of Africa. (20 marks)
5. Citing examples, discuss the characteristics of African industries. (20 marks)
6. "The landscape of Africa is a consequence of physiographic processes". Discuss. (20 marks)



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UNIVERSITY EXAMINATIONS 2018/2019

**FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF SCIENCE IN GEOGRAPHY
AND NATURAL RESOURCES MANAGEMENT WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

NGE 411: LAND AND SOIL RESOURCES MANAGEMENT

Date: 17th September, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 411 LAND AND SOIL RESOURCES MANAGEMENT

1. a) Explain any five (5) management approaches used in natural resources.
(10 marks)

b) Describe soil catena and discuss its relevance to land management.
(20 marks)
2. Examine the land use types applied in land resources management in Kenya.
(20 marks)
3. "Anthropogenic activities are to blame for soil erosion phenomenon". Discuss.
(20 marks)
4. Discuss the causes of land degradation in Africa and suggest possible solutions.
(20 marks)
5. Discuss the relevance of the concept of sustainable development to land and soil resources management.
(20 marks)
6. Write explanatory notes on the following:
 - a) Land valuation (5 marks)
 - b) Salinization (5 marks)
 - c) Hardpan development (5 marks)
 - d) Soil profile (5 marks)



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**FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF SCIENCE IN GEOGRAPHY
AND NATURAL RESOURCES MANAGEMENT WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

NGE 412: BIOLOGICAL RESOURCES MANAGEMENT

Date: 20th September, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 412 BIOLOGICAL RESOURCES MANAGEMENT

1. a) Outline the biological resources. (10 marks)
b) Discuss the concept of Sustainable Development in relation to biological resources management. (20 marks)
2. "Biological resources are to blame for their own vulnerability". Discuss. (20 marks)
3. Discuss the policies, institutional and legal frameworks in the conservation and management of biological resources. (20 marks)
4. Explain how ethical approaches can enhance wildlife conservation and management. (20 marks)
5. Explain the relevance of ecological principles to the conservation and management of biological resources. (20 marks)
6. "Biological resources and development have intricate relationship". Discuss. (20 marks)



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**FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF SCIENCE IN GEOGRAPHY
AND NATURAL RESOURCES MANAGEMENT WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

NGE 417: INDUSTRIAL GEOGRAPHY

Date: 25th September, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Sketch maps and diagrams should be used whenever appropriate



NGE 417 INDUSTRIAL GEOGRAPHY

1. a) Discuss the model of cumulative causation according to Gunnar Myrdal (1957) on the spatial concentration of industries. (20 marks)
b) Describe the main features of International Standard Industrial Classification (ISIC) of economic activities. (10 marks)
2. Describe the Konratiev cycles of industrial development. (20 marks)
3. Evaluate Alfred Weber's Least Cost Locational model. (20 marks)
4. Explain the advantages and disadvantages of Multinational Corporations (MNC) inward investment for host regions in relation to:
 - a) Local economy (4 marks)
 - b) Local firms (4 marks)
 - c) Local community (4 marks)
 - d) Employment (4 marks)
 - e) Political implications (4 marks)
5. Discuss the factors impeding industrial development in Kenya and suggest possible solutions. (20 marks)
6. Write explanatory notes on the following:
 - a) Comparative advantage (4 marks)
 - b) Agglomeration economies (4 marks)
 - c) Division of labour (4 marks)
 - d) Scope of industrial geography (4 marks)
 - e) Characteristics of manufacturing industries (4 marks)